Project 2: Hospital operations with RAG and KG

Concept

Build a knowledge graph that models the entities and relationships within a hospital environment, including departments, staff (doctors, nurses, specialists), equipment, patient flow, bed availability, and common procedures. Integrate this KG with a RAG component that can answer natural language queries about hospital operations.

Healthcare Value

- **Operational Efficiency:** Optimize resource allocation, manage patient flow, and reduce wait times.
- **Staff Management:** Improve scheduling, identify bottlenecks in staff availability, and support emergency response.
- **Cost Reduction:** Pinpoint inefficiencies and areas for improvement in resource utilization.

Key Features for Demo

- **Nested Queries:**
  - "Which operating theaters are available for a orthopedic surgery requiring an Anesthesiologist and a specific type of X-ray machine, and what is the average waiting time for such a surgery in the next 24 hours?" (Requires querying ORs, staff availability, equipment, and historical wait times).
  - "Identify all nurses who are certified in critical care, have worked more than 10 night shifts in the last month, and are available for an ICU shift in the next 8 hours."
- **Discovery of Not-So-Easy-to-Find Patterns:**
  - Identify patterns of equipment failure linked to specific hospital departments or staff shifts, even if formal incident reports are fragmented. (e.g., "Why is X-ray machine A consistently breaking down when operated by technician Y on the night shift?").
  - Uncover hidden dependencies between different hospital resources that lead to unexpected delays (e.g., "Which combination of patient admissions and surgical procedures consistently leads to ICU bed shortages in the afternoon?").

Suggested approach

Synthetic dataset provided. It contains the patterns in the data to meaningfully answer

"Which operating theaters are available for an **orthopedic surgery** requiring an **Anesthesiologist** and a specific type of **X-ray machine**, and what is the average waiting time for such a surgery in the next **24 hours**?"

- "orthopedic surgery," "Anesthesiologist," "X-ray machine" (semantic search to interpret roles/equipment, or direct KG query for these entities).
- "available," "average waiting time," "next 24 hours" (KG traversal for real-time status, historical data, and time-based constraints).

Hybrid: Use semantic search to understand the intent and identify entities, then use these entities to perform precise KG traversals for availability, resource matching, and temporal filtering.

Datasets

Generate synthetic data for the above using your favourite AI Chatbot such as Claude4.

Tip: copy paste the entire text above into your favourite AI chatbot such as Claude4 and ask it to create python script to generate the dataset that exhibits the patterns needed to answer above questions. Ask it to also validate the patterns in the script.

A reference dataset is provided for you in the project2-dataset folder

Dataset details

Total procedures simulated: 506

Total operational logs: 6

Pattern 1: Operating Theater Availability

✓ VALIDATED: At least one Orthopedic Surgery scenario was successfully assigned all required staff and equipment.

Example: Patient PAT000 on 2025-01-01

Staff: {'Anesthesiologist': 'S003', 'Surgeon': 'S016', 'Surgical Nurse': 'S075'}

Equipment: {'X-ray Machine': 'E007', 'Anesthesia Machine': 'E010'}

Pattern 2: Nurse Availability (Critical Care RN, Night Shifts)

✓ VALIDATED: Sample Critical Care Nurse: Walter Powers (ID: S080) has 1 certifications including 'Critical Care RN'.

They have 38 night shifts in their schedule.

Pattern 3: Equipment Failure Linked to Technician/Shift

✓ VALIDATED: Found 3 instances of XRay_Machine_Pattern_001 breaking down.

Latest breakdown: 2025-05-22T22:52:01

Pattern 4: ICU Bottleneck

✓ VALIDATED: At least one ICU bed shortage scenario was detected.

Found 4 delayed ICU admissions.